Abnormality Classification in Gastrointestinal Tract through Endoscopic Imagery using Deep Learning

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INTRODUCTION

Gastrointestinal Tract

- The GI tract is the pathway food takes from your mouth, through the esophagus, stomach, small and large intestine

Endoscopy

- Endoscopy is one of these procedures, it is often used to examine the GI tract embedding a tube with a camera within it

Motivation

- Nearly two million patients die of gastrointestinal diseases worldwide. Image endoscopy is one of the latest technologies in the medical imaging field for the diagnosis of gastrointestinal diseases, such as stomach ulcers, bleeding, and polyps
- The novelty of the proposed methodology lies in developing a system for diagnosis of gastrointestinal diseases

Literature review

SL .No	Title & Referenced Authors	Problem Statement	Algorithms & Techniques used	Datasets	Metrics	Limitations
1.	EndoL2H: Deep Super-Resolution for Capsule Endoscopy Yasin Almalioglu , Graduate Student Member, IEEE, Kutsev. December 2020	To improve the low resolution image to high resolution image to improve the visual of the image	Spatial Attention-Based Super Resolution cGAN High Fidelity Loss Function	The dataset was divided into three subsets: 17, 220 images for training, 2, 000 images for validation, and 2, 000 for testing	PSNR, SSIM, LPIPS, and GMSD The result image was sharper and more informative than before. The image was clear and sharper according various standards.	This only converts low resolution image to high resolution image.
2.	Registration of Consecutive Frames From Wireless Capsule Endoscopy for 3D Motion Estimation , IEEE PAPER Marina Oliveira ,Helder Araujo , (Member, Ieee) August 2021	To capture the image from the capsule frame by frame. To estimate the trajectory for the capsule to reconstruct the path for easy detection of abnormalities.	Direct (pixel-based) , feature based, artificial intelligence strategies.	The data used for this assessment is the Mirocam dataset obtained with our novel experimental setup	MEIR, MPIR, SIFT, SURF, MSER, and LIOP The resulted image was cross checked with manual data and its almost same and visually correct.	The detection of the travel track of capsule was hard. This only captures and predicts the photo for the data.

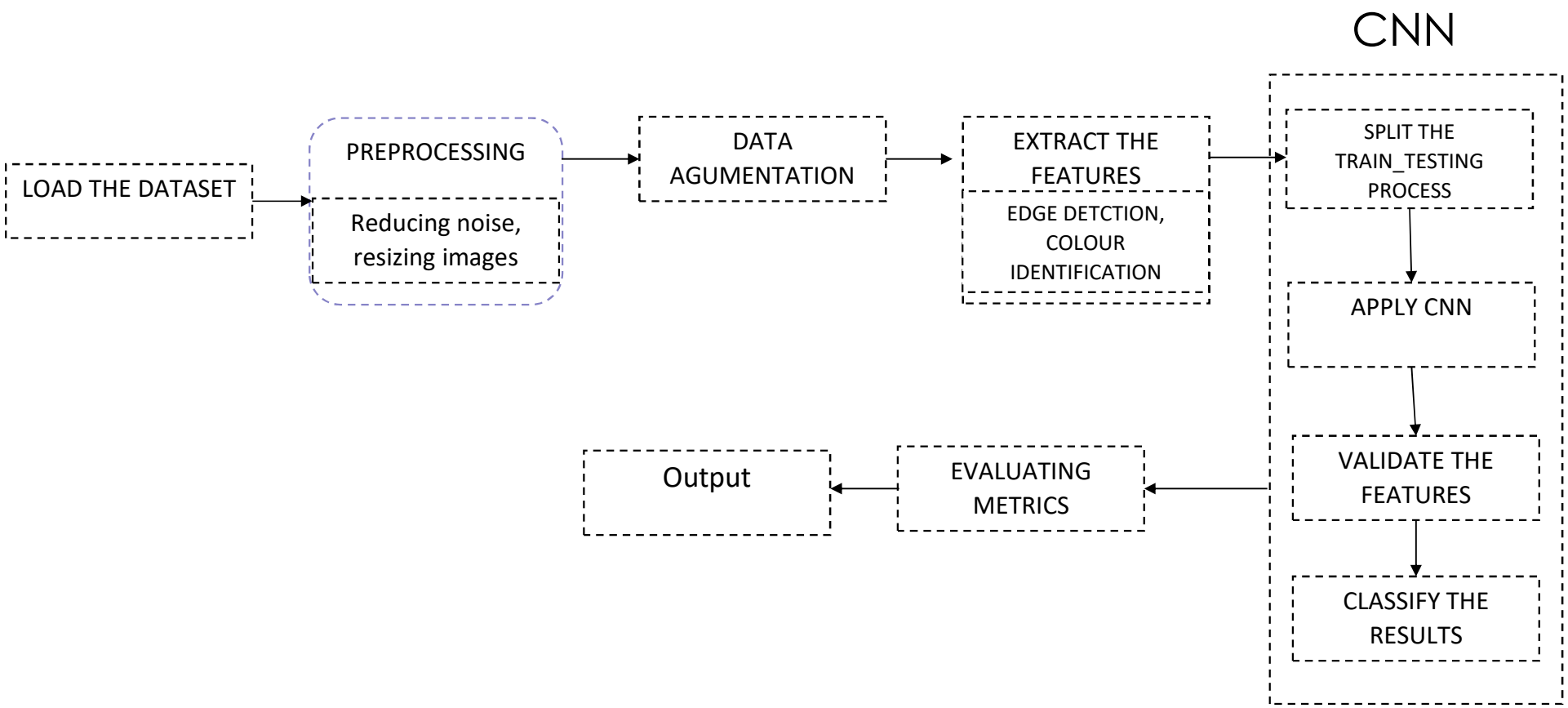
3.	A Learning Based Microultrasound System for the Detection of Inflammation of the Gastrointestinal Tract IEEE TRANS MED IMAGING Shufan yang , christina lemke January 2021	To detect inflammation in GI tract	Inception V3	B-Scan Generation US Data samples	The Inception V3 model used several techniques for optimizing the network for better model adaptation. It is computationally less expensive. It uses auxiliary	the requirement of more data to overcome possible overfitting and more computational resource. The fundamental way to solve these 2 problem for computer vision system is to use sparsely connected architecture to replace the fully connected one
4.	Improving Low-Resolution Image Classification by Super-Resolution with Enhancing High-Frequency Content , IEEE PAPER Liguo Zhou, Guang Chen, Mingyue Feng 2020	Improving Low-Resolution Image Classification by Super-Resolution with Enhancing High-Frequency Content	Convelutional neural network	Endoscopy video review	Compared with other state-of-the-art super-resolution methods, the experiment results show that our method can recover more high frequency contents in high-resolution image reconstructing	Only improves the image quality

5.	Informative frame classification for endoscopy video SCIENCE DIRECT By Jung Hwan Oh 2010	Aims to classify unwanted images from the endoscopic test video	edge-based and clustering-based	Endoscopic test video.	two informative frame classification techniques are greater than 90% and 95%	This method only gives us the informative image and does not classify abnormalities.
6.	Color-based template selection for detection of gastric abnormalities in video endoscopy SCIENCE DIRECT By Hussam Ali Febuary 2020	Aims to predict gastric abnormalities using colour detection.	Color-based segmentation method has been applied for abnormal area selection.	Endoscopic test videos.	method achieves 86.6% accuracy	This detects based on colour differences. Miss classification of abnormalities is a possibility

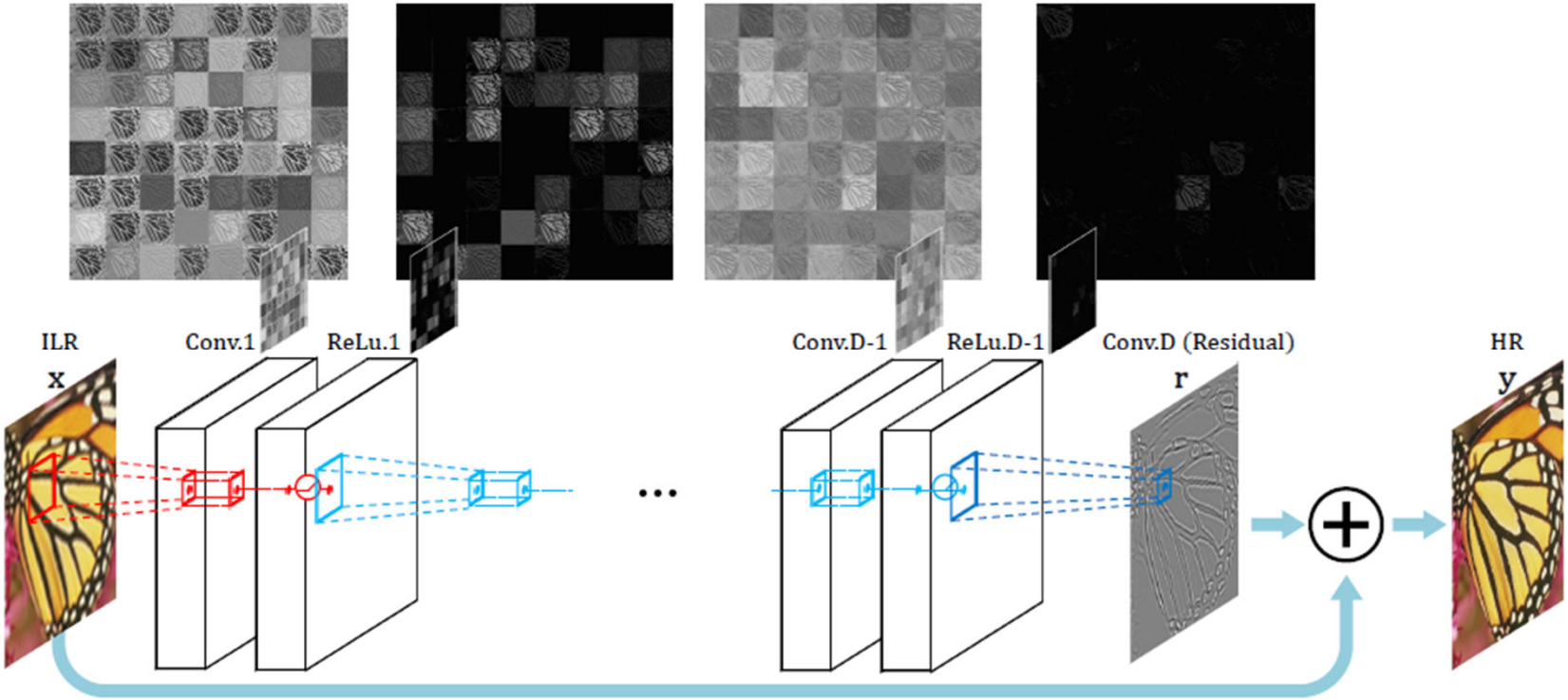
Existing System

- **Input data:** The dataset endoscopic test are given to machine.
- **Dataset:**
  - K-vasir dataset contains 3000 images
  - 80% of the dataset is divided for training and 20% for validation
- **Task at hand:** Low resolution image is mapped to high resolution images
- **Algorithm used**: Super resolution algorithms

Architecture of Existing System

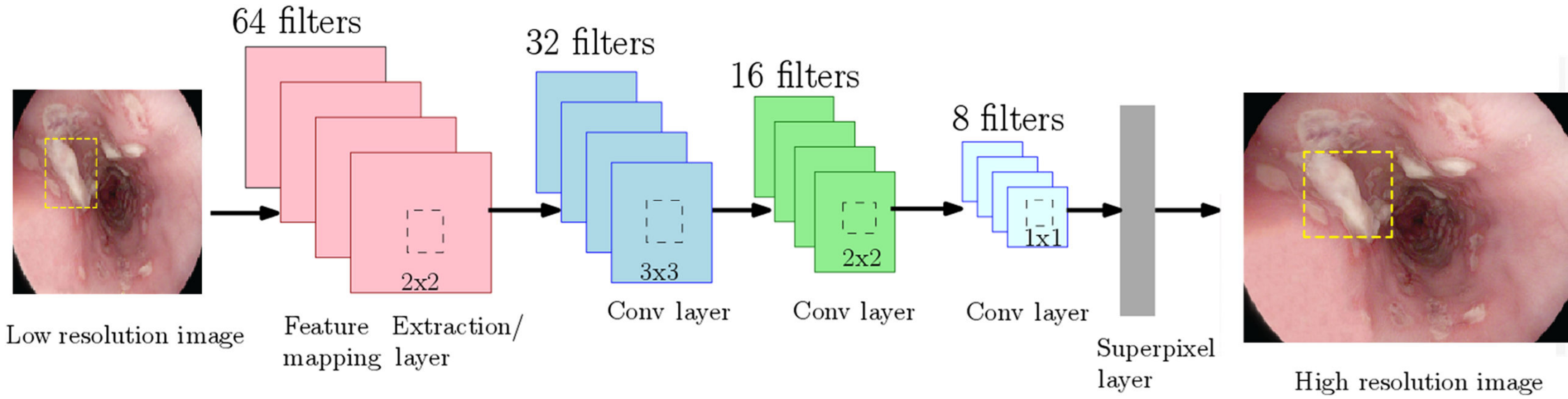


Super-Resolution Working



Super Resolution Algorithm

- The low-resolution images are divided into three separate images with three RGB channels.
- Then the three images are trained by a convolutional neural network to obtain three output images.
- Finally, the obtained three images are fused to obtain the final high-resolution image reconstruction.
- More colour features and high frequency information can be obtained by using RGB multi-channel as the input image, which is more effective in SR reconstruction.
- The evaluation indexes such as the peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) are measured.


64 filters
32 filters
16 filters
8 filters
2x2
3x3
2x2
1x1
Low resolution image
Feature Extraction/
mapping layer
Conv layer
Conv layer
Conv layer
Superpixel
layer
High resolution image

Metrics

- Precision = TP/(TP+FP)
- The precision value lies between 0 and 1.
- Recall = TP/(TP+FN)
- **True Positive (TP)** — model correctly predicts the positive class (prediction and actual both are positive).
- **True Negative (TN)** — model correctly predicts the negative class (prediction and actual both are negative).
- **False Positive (FP)** — model gives the wrong prediction of the negative class (predicted-positive, actual-negative).
- **False Negative (FN)** — model wrongly predicts the positive class (predicted-negative, actual-positive).
- F1 score gives the same weightage to recall and precision.

$$F1\ score = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} = \frac{2 * (Precision * Recall)}{(Precision + Recall)}$$

Limitation Of Existing System

- This model only changes the resolution of the image from low to high
- The detection of abnormalities is done manually
- The detection of abnormalities takes lots of concentration, time and effort to diagnose the disease
- Even when the image is converted into high resolution picture there are chances of misdiagnosing

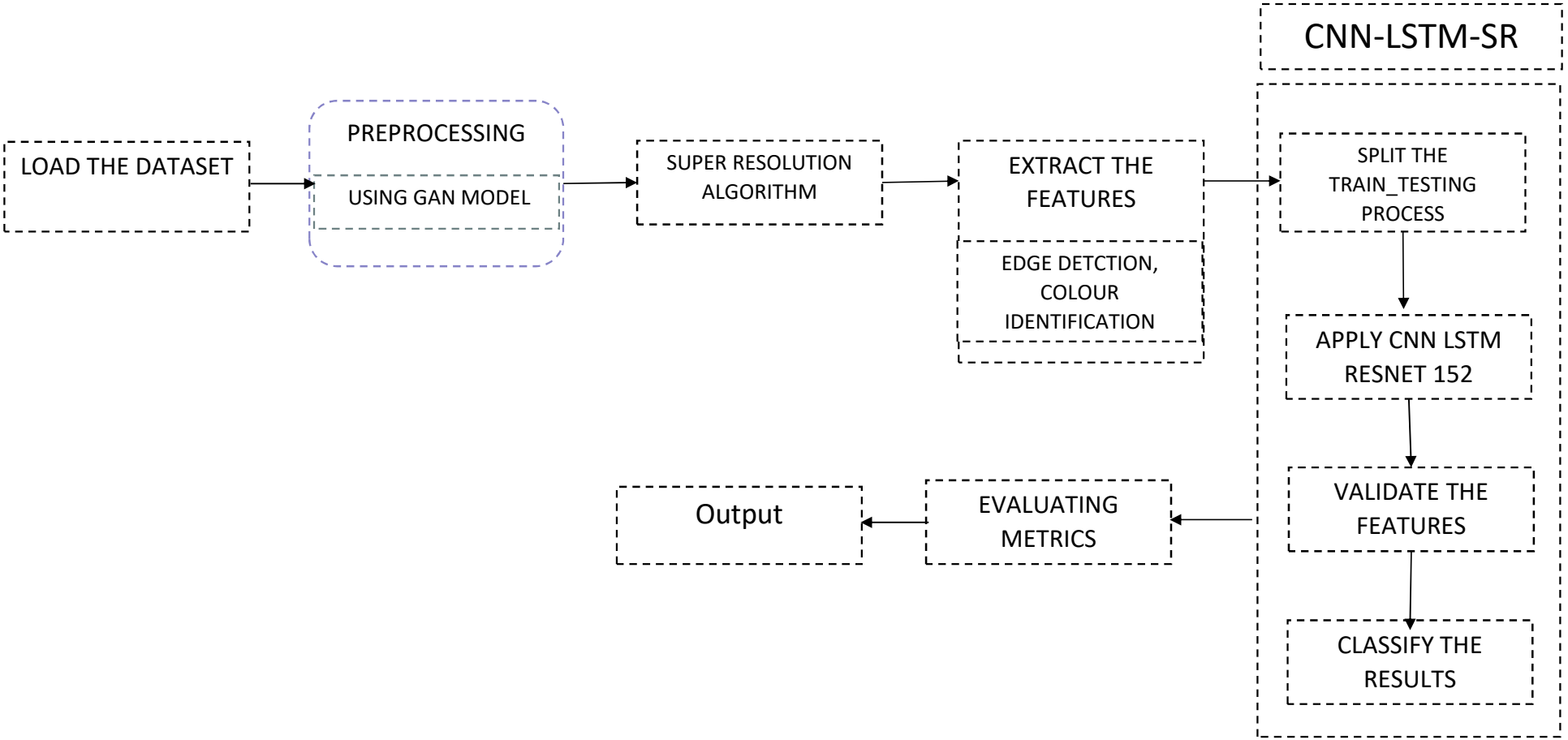
PROBLEM DEFINITION

- Esophageal, stomach, and colorectal cancers account for 2.8 million new cases and 1.8 million deaths worldwide per year.
- Due to its complex nature of gastrointestinal infections, gastroscopy instruments are not suitable for identifying and examining gastrointestinal infections.
- Our study helps doctors to identify these gastrointestinal infections with the help of our deep learning model which uses CNN-LSTM hybrid model.

OBJECTIVE OF THE PROPOSED SYSTEM

- To convert low resolution images to high resolution and classify the abnormalities in endoscopic images into different classes using CNN-LSTM-SR

Architecture Of Proposed System



Module Description

Pre-Processing

- We resize all image into same size to train our system. All image should be of same size for better training.
- Filtering important images from the dataset
- We remove unwanted noises from the image.

Gan model

- A category-guided feature generation (CFG) module is proposed to generate the refined global and local category-guided features for the segmentation. The GAN model algorithm is used to generate the global and local category-guided features.

Feature Extraction

- Feature is extracted by CNN-LSTM model
- Some of the features are edge detection, Gray scaling, brightness, sharpness etc.

Splitting Of the Dataset

- The dataset in splited into:
  - 80% training dataset
  - 20% testing dataset

Applying CNN LSTM resnet152

- **Convolution:** The purpose of convolution is to extract features from the input image. It preserves the spatial relationship between pixels by learning image features using small squares of input data. It is usually followed by Relu.
- **Relu:** It is an element-wise operation that replaces all negative pixel values in the feature map by zero. Its purpose is to introduce non-linearity in a convolution network.
- **Pooling:** Pooling (also called downsampling) reduces the dimensionality of each feature map but retains important data.
- **LSTM**: Long short-term memory is an artificial neural network used in the fields of artificial intelligence and deep learning. Unlike standard feedforward neural networks, LSTM has feedback connections. Such a recurrent neural network can process not only single data points, but also entire sequences of data.
- **Fully-connected layer:** It is a multilayer perceptron that uses the softmax function in the output layer. Its purpose is to use features from previous layers for classifying the input image into various classes based on training data.
- **ResNet152**: Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth.

Classification

- Lesion location mining method is used for classification in CNN-LSTM with RESNET 152 model
- The classification labels are:
  - dyed-lifted-polyps
  - dyed-resection-margins
  - Esophagitis
  - normal-z-line
  - polyps
  - ulcerative-colitis

Metrics

- Precision = TP/(TP+FP)
- The precision value lies between 0 and 1.
- Recall = TP/(TP+FN)
- **True Positive (TP)** — model correctly predicts the positive class (prediction and actual both are positive).
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- F1 score gives the same weightage to recall and precision.

$$F1\ score = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} = \frac{2 * (Precision * Recall)}{(Precision + Recall)}$$

Input and output

- Input:
  - The continuous picture of an endoscopic test is given to the machine
  - 3000 images of GI tract with abnormalities
  - This dataset is used from kaggle
- Output:
  - Labelled pictures in a confusion matrix

Hardware and software requirements

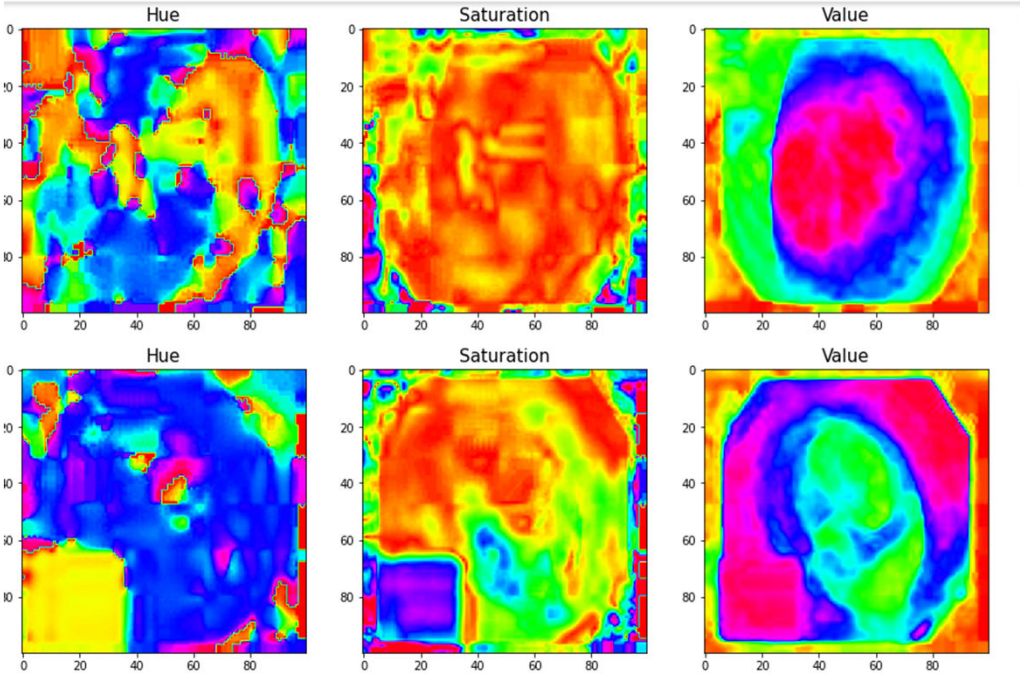
- Hardware requirements:
- RAM: Minimum 4GB
- ROM:256 GB
- Software requirements:
- Operating System: 64 Bit(Windows 8/10/11)
- Platform : Python , Google Colab , jupyter notebook

Implementation Of Existing System

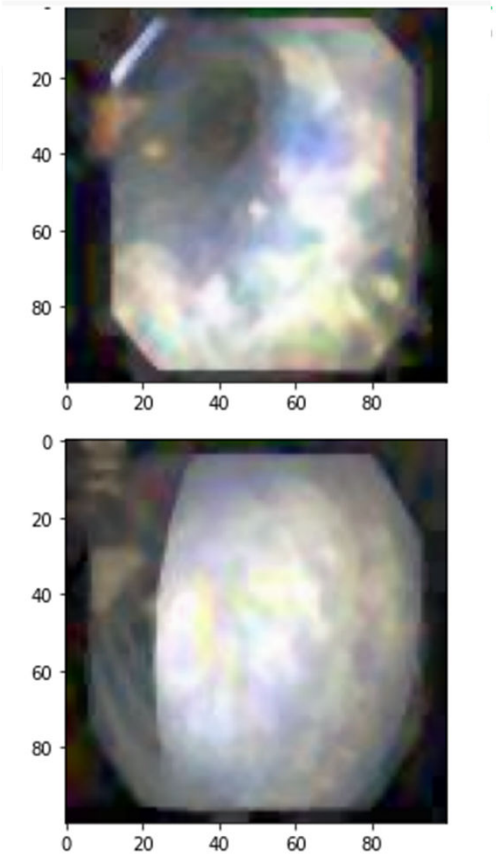
- Loading Dataset

```
[5] #change the location here
    train_dir = '/content/drive/MyDrive/SR_Endo/dataset'
    labels= ['dyed-lifted-polyps', 'dyed-resection-margins', 'esophagitis', 'normal-z-line', 'polyps', 'ulcerative-colitis']# dataset_name

    test_size=0.85
    for sp in labels:
        print('{} {} images'.format(sp, len(os.listdir(os.path.join(train_dir, sp)))))
    train = []
    for species_num, sp in enumerate(labels):
        for file in os.listdir(os.path.join(train_dir, sp)):
            #change the location here
            train.append(['/content/drive/MyDrive/SR_Endo/dataset//{}//{}'.format(sp, file), species_num, sp])

    train = pd.DataFrame(train, columns=['file', 'species_num', 'species'])

    print('Training Data: ',train.shape)
```

```
dyed-lifted-polyps 200 images
dyed-resection-margins 200 images
esophagitis 200 images
normal-z-line 200 images
polyps 200 images
ulcerative-colitis 200 images
Training Data:  (1200, 3)
```

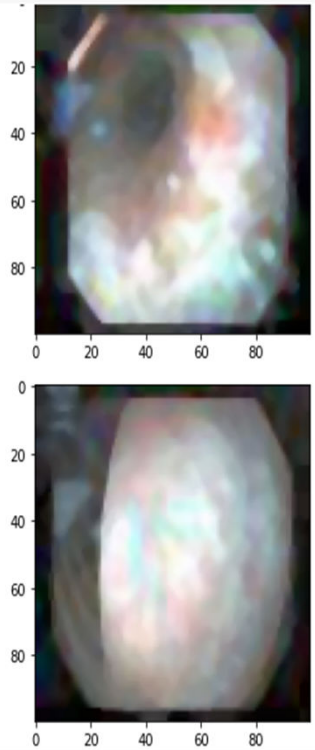


HSV SRGAN



SRGAN

```
[11] for i in train['file']:
         im1 = Image.open(i)
         im2 = im1.filter(ImageFilter.MedianFilter(size = 3))
         im2.save(i)
         plt.imshow(im2)
         plt.show()
```



Super Resolution Image

[24]
```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.legend(['training','validation'])
plt.title('Training and validation accuracy')
plt.xlabel('epoch')
```

```
Text(0.5, 0, 'epoch')
```

[23]
```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.legend(['training','validation'])
plt.title('Training and validation Loss')
plt.xlabel('epoch')
```

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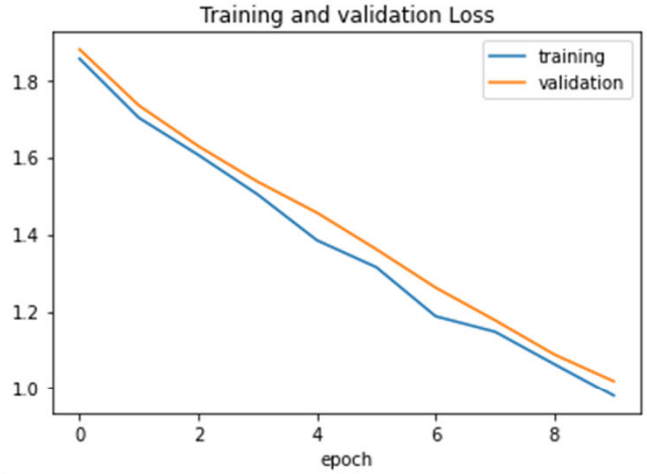
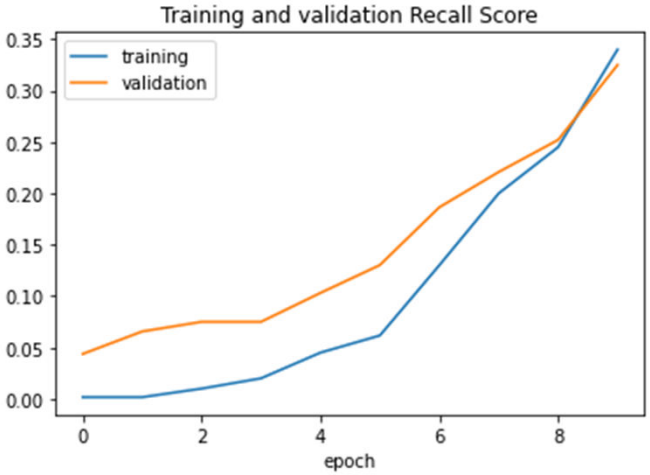


[26]

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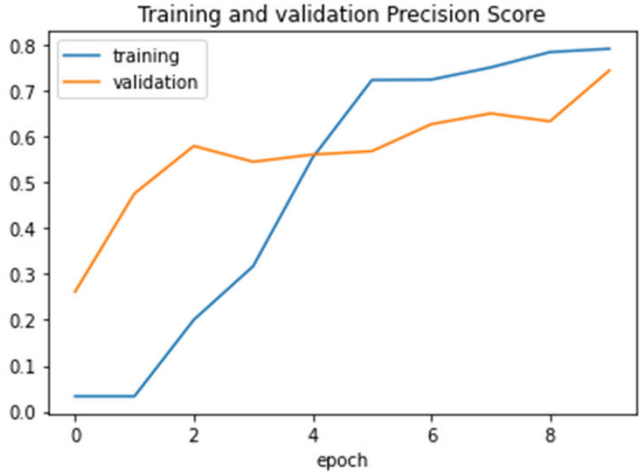

Implementation of proposed work

Loading Dataset

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[17] #change the location here
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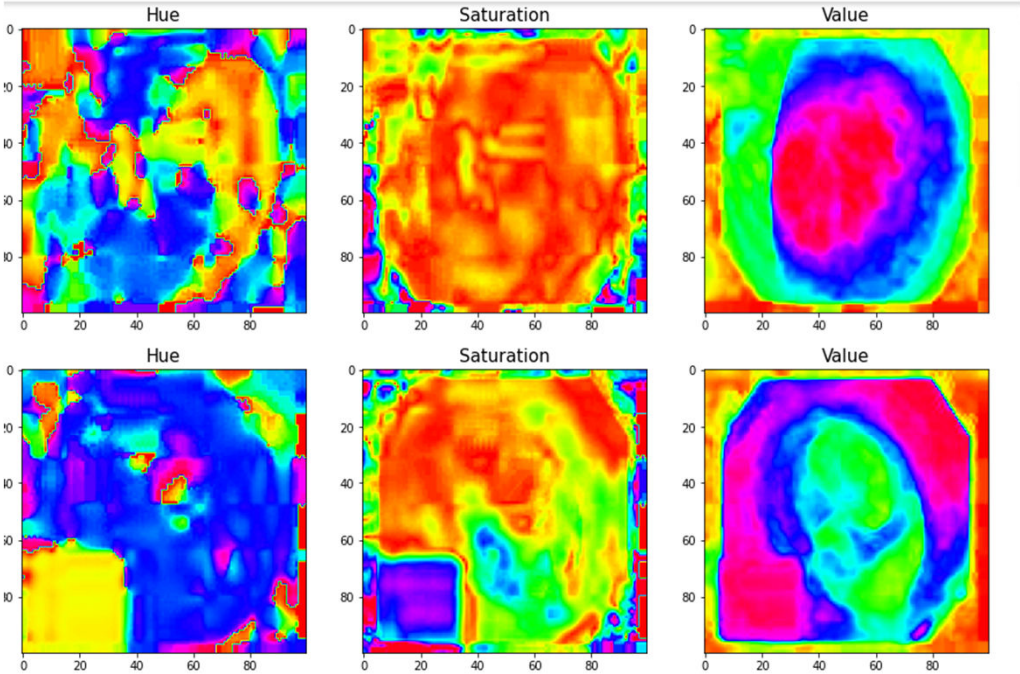

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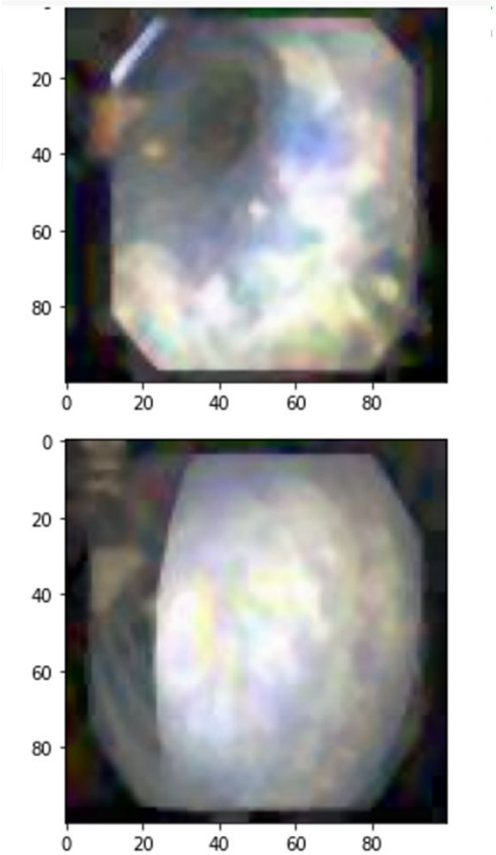

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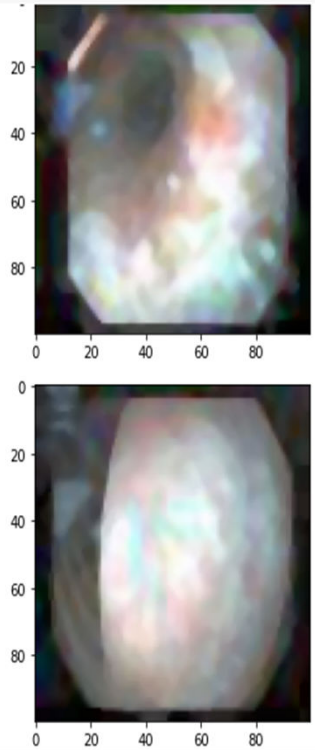


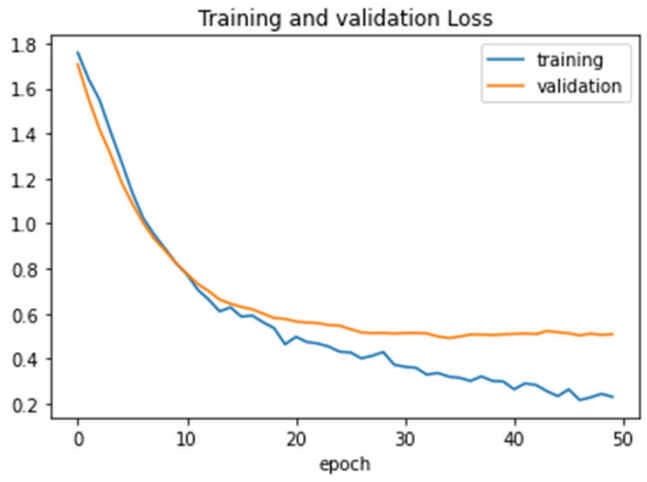
Super Resolution Image

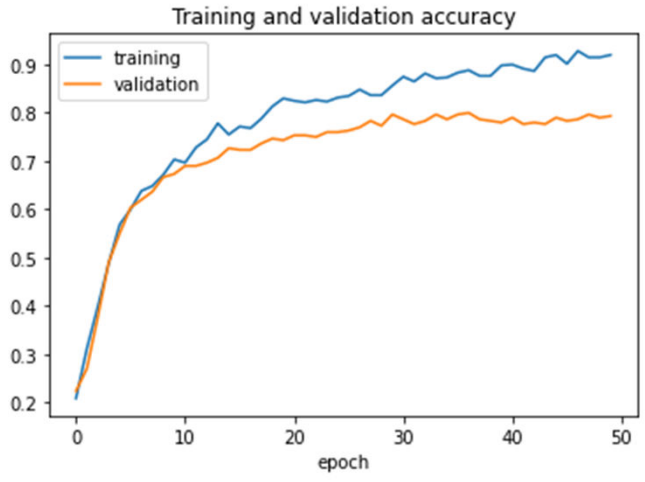
Classification report

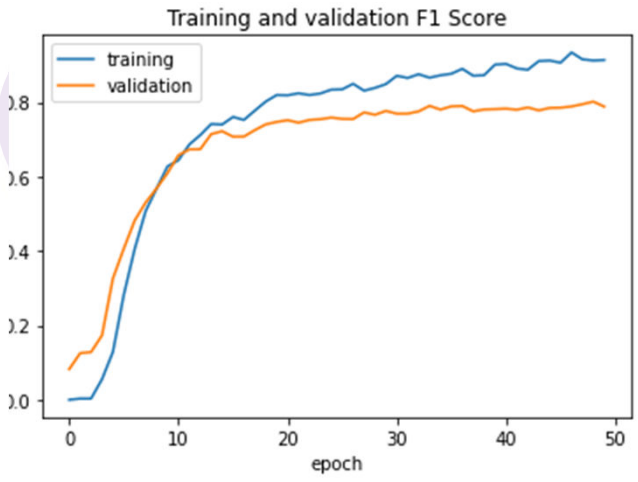
	precision	recall	f1-score	support
0	0.91	0.82	0.86	50
1	0.85	0.90	0.87	50
2	0.85	0.78	0.81	50
3	0.80	0.86	0.83	50
4	0.92	0.98	0.95	50
5	0.96	0.94	0.95	50
accuracy			0.88	300
macro avg	0.88	0.88	0.88	300
weighted avg	0.88	0.88	0.88	300

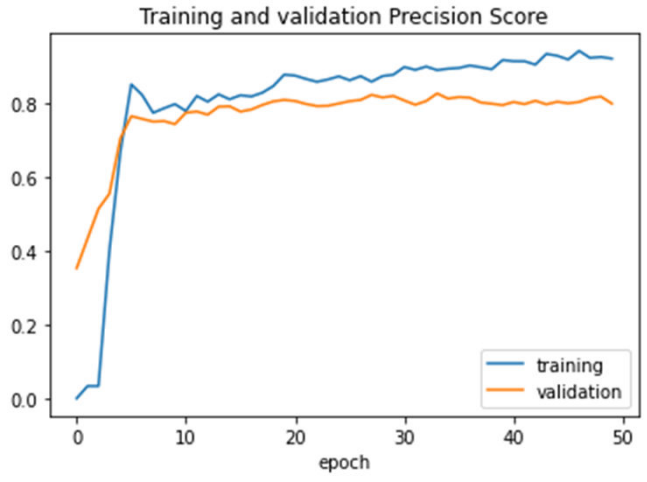
Comparison Table

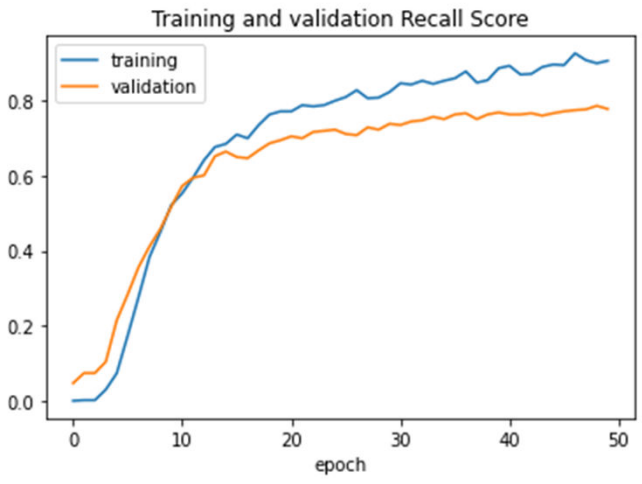
	CNN	RNN
Accuracy	0.779167	0.880000
F1_score	0.859023	0.879553
Precision	0.860822	0.881334
recall	0.859167	0.880000
sensitivity	0.829146	0.918367
specificity	0.915000	0.836735

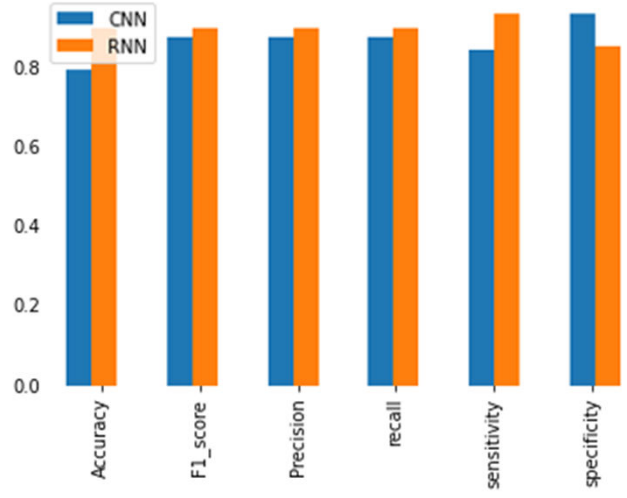

Training and validation Loss
training
validation
epoch


Training and validation accuracy
training
validation
epoch


Training and validation F1 Score
training
validation
epoch


Training and validation Precision Score
training
validation
epoch


Training and validation Recall Score
training
validation
epoch


CNN
RNN
Accuracy
F1_score
Precision
recall
sensitivity
specificity

PLAN OF ACTION

CONCLUSION

- Gastrointestinal tract abnormalities are very hard to notice and easy to misdiagnose. The proposed system helps us in identifying the abnormalities and give a universal diagnose. This save much times and effort for the specialist instead of manual diagnose.

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THANK YOU